



chapter6a_moments...

Chapter 6A: Moments, Skewness, and Kurtosis
MATH/STAT 394: Probability I

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Learning Goals

By the end of this lecture, students should be able to:

- define and interpret the mean, median, and mode;
- define moments, standardized moments, and standardized moments;
- define skewness and kurtosis;
- understand symmetry of distributions;
- define sample moments;
- compute the mean and variance of the sample mean.

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Summaries of a Distribution

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Why Summaries Matter

A probability distribution contains a large amount of information. Often we want simpler numerical summaries:

- Where is the center?
- How spread out is the distribution?
- Is it symmetric?
- Are extreme values common?

Main idea
Moments provide numerical summaries of distributions.

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Measures of Standardized Tendency

Three common summaries of the center are:

- Mean (expected value)
- Median
- Mode

These summarize different aspects of a distribution.

Important
Different notions of "center" can give very different values.

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Median

Definition
A number m is a median of X if

$$P(X \leq m) \geq \frac{1}{2}$$
$$P(X \geq m) \geq \frac{1}{2}$$

Intuitively
The graph of f_X is a mirror image across the vertical line $x = m$:
The area under f_X to the left of m is the same as the area to the right of m .
Median is less sensitive to extreme values than means.

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Mode

Definition
A value c is called a mode of a random variable X if it maximizes

$$f_X(c)$$

for a discrete random variable with PMF $f_X(c)$.
Equivalently,
for a continuous random variable with PDF $f_X(c)$.

$$f_X(c) \geq f_X(x) \quad \forall x$$
$$f_X(c) \geq f_X(x) \quad \forall x$$

Interpretation
The mode is the most likely value (discrete case) or the point of highest density (continuous case).

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Mean vs Median

Suppose salaries are: 40, 42, 45, 46, 48, 50, 50, 1000.
Then mean = 46,
while median = 181.6.

Observation
The mean is heavily affected by extreme values (outliers), while the median is robust.

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Best Prediction Depends on Loss Function

Suppose we predict an unknown random variable X by a constant c .
Two common error measures:

$$E[(X - c)^2]$$

(mean squared error)
and
 $E[|X - c|]$
(mean absolute error).

Key fact
The mean minimizes mean squared error.
A median minimizes mean absolute error.

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Mean and Variance Are Not Enough

Two distributions can have: same mean, same variance,
yet look very different.

Possible differences:
asymmetry;
heavy tails;
sharp peaks;
multiple modes.

Conclusion
Higher moments help describe these additional features.

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Moments and Symmetry

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Kinds of Population Moments

Definitions
The n th population moment:
$$\mu_n' = E(X^n)$$

The n th central population moment:
$$\mu_n = E[(X - \mu)^n]$$

The n th standardized population moment:
$$\mu_n'' = \frac{\mu_n}{\sigma^n} = E\left[\left(\frac{X - \mu}{\sigma}\right)^n\right]$$

So, we have the following relations
Mean is the 1st population moment $\mu_1' = \mu$, determines center
Variance is the 2nd standardized population moment $\mu_2'' = \sigma^2$, determines spread
Skewness is the 3rd standardized population moment μ_3'' , determines asymmetry
Kurtosis is the 4th standardized population moment μ_4'' , determines tail behavior

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Symmetry About a Point

Definition
A random variable X is symmetric about μ if

$$X - \mu \stackrel{d}{=} \mu - X = -(X - \mu),$$

where $\stackrel{d}{=}$ denotes equality in distribution.

Let $Y = X - \mu$.
Then the CDF of Y is

$$F_Y(y) = P(Y \leq y) = P(X - \mu \leq y) = P(X \leq y + \mu) = F_X(y + \mu).$$

If X has PDF f_X , then differentiating gives

$$f_Y(y) = f_X(y + \mu).$$

Similarly, for $-Y = \mu - X$,

$$f_{-Y}(y) = f_X(\mu - y) = f_X(\mu - y).$$

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Symmetry About a Point (Continue)

Therefore, X is symmetric about μ when

$$f_X(\mu + y) = f_X(\mu - y) \quad \text{for all } y.$$

In this example, values far to the left (from μ , the center) act "better" compared to the values far to the right.
The graph of f_X is a mirror image across the vertical line $x = \mu$.
If the mean exists, then the center of symmetry is the mean:
 $E[X] = \mu$.

More generally, all odd central moments that exist are zero:
$$E[(X - \mu)^{2k+1}] = 0.$$

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Symmetry and Odd Central Moments

Suppose X is symmetric about μ . Then $f_X(\mu + t) = f_X(\mu - t)$.
Consider an odd central moment, say the $(2k+1)$ st:

$$\mu_{2k+1} = E[(X - \mu)^{2k+1}] = \int_{-\infty}^{\infty} (x - \mu)^{2k+1} f_X(x) dx.$$

Make the change of variable $t = x - \mu$.
Then $x = \mu + t$, so

$$\mu_{2k+1} = \int_{-\infty}^{\infty} t^{2k+1} f_X(\mu + t) dt.$$

Now define $g(t) = t^{2k+1} f_X(\mu + t)$.
Then $g(-t) = (-t)^{2k+1} f_X(\mu - t) = -t^{2k+1} f_X(\mu + t) = -g(t)$.
Thus g is an odd function. Therefore,
$$\int_{-\infty}^{\infty} g(t) dt = 0.$$

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A second way of proving it...

Proposition
Let X be a random variable symmetric about μ . Then, for every integer $k \geq 0$,
$$E[(X - \mu)^{2k+1}] = 0,$$

provided the moment exists.

Proof.
Let $Y = X - \mu$. By symmetry, $Y \stackrel{d}{=} -Y$. Therefore, $E[Y^{2k+1}] = E[(-Y)^{2k+1}] = E[-Y^{2k+1}] = -E[Y^{2k+1}]$. But since $2k+1$ is odd,
 $E[Y^{2k+1}] = -E[Y^{2k+1}]$. Hence, $E[Y^{2k+1}] = 0$. So, $E[(X - \mu)^{2k+1}] = 0$. Since $Y = X - \mu$, $E[(X - \mu)^{2k+1}] = 0$.

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Examples of Symmetry

A distribution is symmetric about μ when points equally far from μ have the same probability density.

- Normal distribution: If $X \sim N(\mu, \sigma^2)$, then $f_X(\mu + t) = f_X(\mu - t)$.

For any t ,
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$f_X(\mu + t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right) = f_X(\mu - t)$$

Thus, the normal distribution is symmetric about μ .
Uniform distribution on a symmetric interval: If $X \sim \text{Unif}(a, b)$ with $a = -\lambda$ and $b = \lambda$, then
$$f_X(x) = \frac{1}{2\lambda} \quad \text{for } -\lambda \leq x \leq \lambda, \text{ otherwise } 0.$$

Hence, for every t , $f_X(\mu + t) = f_X(\mu - t)$, since either both points lie inside the interval and have density $\frac{1}{2\lambda}$, or both lie outside and have density 0. Hence, $E[Y^{2k+1}] = 0$. So, $E[(X - \mu)^{2k+1}] = 0$.
Binomial distribution with $p = \frac{1}{2}$: If $X \sim \text{Bin}(n, \frac{1}{2})$, then
$$P(X = k) = \binom{n}{k} \left(\frac{1}{2}\right)^n = \binom{n}{n-k} \left(\frac{1}{2}\right)^n = P(X = n - k).$$

Thus, the distribution is symmetric about $\frac{n}{2}$.

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Skewness

Definition
The skewness of X is

$$\text{Skew}(X) = E\left[\left(\frac{X - \mu}{\sigma}\right)^3\right].$$

Interpretation:
positive skewness: long right tail;
negative skewness: long left tail;
zero skewness: often indicates symmetry.

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Examples of Skewness

Exponential distributions are right-skewed.
Income distributions are often right-skewed.
Normal distributions have skewness 0.

Important
Zero skewness does not necessarily imply symmetry.

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Kurtosis

Definition
The kurtosis of X is

$$\text{Kurt}(X) = E\left[\left(\frac{X - \mu}{\sigma}\right)^4\right].$$

Kurtosis measures tail heaviness and extremeness.
 $\text{Kurt}(N(0,1)) = 3$. If $\text{Kurt}(X) > 3$, then X has heavier/lighter tails compared to normal.

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Common Discrete Distributions

Distribution	Mean	Median	Mode	Variance	Skewness	Kurtosis
Bernoulli(p)	p	$\begin{cases} 0, & p < 1/2 \\ 1/2, & p = 1/2 \\ 1, & p > 1/2 \end{cases}$	$\begin{cases} 0, & p < 1/2 \\ 1/2, & p = 1/2 \\ 1, & p > 1/2 \end{cases}$	$p(1-p)$	$\frac{1-2p}{\sqrt{p(1-p)}}$	$\frac{1-6p(1-p)}{p(1-p)^2}$
Binomial(n, p)	np	$\approx np$	$\lfloor (n+1)p \rfloor$	$np(1-p)$	$\frac{1-2p}{\sqrt{np(1-p)}}$	$\frac{1-6p(1-p)}{np(1-p)^2}$
Poisson(λ)	λ	$\begin{cases} 0, & \lambda < 1 \\ 1, & \lambda \geq 1 \end{cases}$	$\begin{cases} 0, & \lambda < 1 \\ 1, & \lambda \geq 1 \end{cases}$	λ	$\frac{1-2\lambda}{\sqrt{\lambda}}$	$\frac{1-6\lambda}{\lambda^2}$

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Common Continuous Distributions

Distribution	Mean	Median	Mode	Variance	Skewness	Kurtosis
Uniform(a, b)	$\frac{a+b}{2}$	$\frac{a+b}{2}$	none	$\frac{(b-a)^2}{12}$	0	$\frac{1}{3}$
Normal(μ, σ^2)	μ	μ	μ	σ^2	0	$\frac{3}{2}$
Exponential(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$	0	$\frac{1}{\lambda^2}$	2	$\frac{9}{2}$
Gamma(k, λ)	$\frac{k}{\lambda}$	$\frac{k-1}{\lambda}$ (no closed form)	0	$\frac{k}{\lambda^2}$	$\frac{3-k}{\sqrt{k}}$	$\frac{3}{2} + \frac{6-k}{k}$
Student(ν)	0	0	0	$\frac{\nu}{\nu-2}$	0	$\frac{3\nu}{\nu-4}$

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Sample Moments

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Sample Moments

Suppose $X_1, \dots, X_n$ are i.i.d. random variables.
The k th sample moment is

$$M_k = \frac{1}{n} \sum_{i=1}^n X_i^k.$$

Sample moments estimate population moments.
The first sample moment, $M_1 = \frac{1}{n} \sum_{i=1}^n X_i$, is an important quantity in Statistics, also known as the Sample Mean, denoted by $\bar{X}$. It estimates the first population moment, i.e., $\mu = E[X]$, the population mean.

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Point-Mass (Degenerate) Distribution

Definition
A random variable X has a point-mass distribution at $c \in \mathbb{R}$ if

$$P(X = c) = 1.$$

Equivalently, $X = c$ almost surely. We may write $X \sim c$.

Its PMF is $p_X(x) = P(X = x) = \begin{cases} 1, & x = c \\ 0, & x \neq c \end{cases}$

Its CDF is $F_X(x) = P(X \leq x) = \begin{cases} 0, & x < c \\ 1, & x \geq c \end{cases}$

Thus, the CDF has a jump of size 1 at $x = c$.
 $F_X(x) = F_X(c^-) = P(X = c) = 1$.

Always Summaries
Since X always equals c ,
 $E[X] = c, E[X^2] = c^2, \dots, E[X^k] = c^k$.

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i.i.d. Random Variables (Random Samples)

Definition: i.i.d. Random Variables
We say that random variables $X_1, \dots, X_n$ are independent and identically distributed (i.i.d.) with distribution F , and write

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$$

if

- $X_1, \dots, X_n$ are independent;
- $X_1, \dots, X_n$ have distribution F .

Terminology: If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$, then we also call a random sample of size n from the population distribution F .
Quantities associated with F are called population quantities, while quantities computed from the random sample are called sample quantities.

Example
If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$, then we also call a random sample of size n from the population distribution F .
Quantities associated with F are called population quantities, while quantities computed from the random sample are called sample quantities.

Suppose the population consists of the n students of all University of Washington undergraduate students.
Now suppose we randomly select n undergraduate students and record their GPAs.
Here,

- each X_i represents one student's GPA;
- all X_i 's have distribution F ;
- the observations are assumed to be independent.

A possible sample (after observing $X_1, \dots, X_n$) is 3.5, 3.7, 4.0, 2.9, 3.3, 3.8, 3.4, 2.7, 3.0, 3.2, 3.6, 3.3.
We would then call these numbers an observation of our random variables. If we perform the experiment and define $X = \text{GPA}$, then, drawing a random sample $X_1, \dots, X_n$ results in a few numbers (observations) such as 3.5, 3.7, 4.0, 2.9, 3.3, 3.8, 3.4, 2.7, 3.0, 3.2, 3.6, 3.3. These are the values of the random variables $X_1, \dots, X_n$. There is our random sample. So we write $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$.
In practice, we usually do not know the population distribution F . We only know the population mean μ and variance σ^2 .
From the population and use the random sample to estimate unknown population quantities.

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Sample Mean and Sample Variance

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$. $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2$.

Definitions
The sample mean and sample variance are

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Mean and Variance of the Sample Mean and Sample Variance
where $\mu = E(X_i) = \mu$.

Proofs (Sample Mean)
Expected Value
By linearity of expectation,
$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

Thus, $\bar{X}$ is an unbiased estimator of μ .

Variance
Since $X_1, \dots, X_n$ are independent,
$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}$$

Thus, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$.

A Useful Identity (for the Sample Variance, before the proofs)
To study S^2 , first rewrite its numerator.

Sum of Squares Identity
$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n(\bar{X})^2$$

Proof: Write $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$.
Therefore,
$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu - (\bar{X} - \mu))^2 = \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu) + n(\bar{X} - \mu)^2$$

But $\sum_{i=1}^n (X_i - \mu) = n(\bar{X} - \mu)$. Hence,
$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Proofs (Sample Variance)
Using the identity from the previous slide, $(n-1)S^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Take expectations:
$$(n-1)E(S^2) = \sum_{i=1}^n E[(X_i - \mu)^2] - nE[(\bar{X} - \mu)^2]$$

Since $E[(X_i - \mu)^2] = \sigma^2$ and $E[(\bar{X} - \mu)^2] = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, we obtain
$$(n-1)E(S^2) = n\sigma^2 - n\left(\frac{\sigma^2}{n}\right) = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$$

Therefore,
$$E(S^2) = \sigma^2$$

Why Do We Divide by $n-1$?
The denominator $n-1$, rather than n , corrects for the downward bias caused by replacing the unknown population mean μ with the sample mean $\bar{X}$.

Proofs (Sample Variance)

Using the identity from the previous slide,
$$(n-1)S^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

Take expectations:
$$(n-1)E(S^2) = \sum_{i=1}^n E[(X_i - \mu)^2] - nE[(\bar{X} - \mu)^2]$$

Since $E[(X_i - \mu)^2] = \sigma^2$ and $E[(\bar{X} - \mu)^2] = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, we obtain
$$(n-1)E(S^2) = n\sigma^2 - n\left(\frac{\sigma^2}{n}\right) = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$$

Therefore,
$$E(S^2) = \sigma^2$$

Proof Setup: Variance of the Sample Variance
Define the centered random variables
 $Y_i = X_i - \mu$.
Then $E(Y_i) = 0$, $E(Y_i^2) = \sigma^2$, $E(Y_i Y_j) = 0$.
Also define $A = \sum_{i=1}^n Y_i^2$, $B = \sum_{i=1}^n Y_i$.
Because $\bar{X} - \mu = \frac{B}{n}$,
the sum-of-squares identity gives
$$(n-1)S^2 = A - \frac{B^2}{n}$$

Therefore,
$$S^2 = \frac{1}{n-1} \left(A - \frac{B^2}{n} \right)$$

Thus,
$$\text{Var}(S^2) = \frac{1}{(n-1)^2} \text{Var}\left(A - \frac{B^2}{n} \right)$$

Equivalently,
$$\text{Var}(S^2) = \frac{1}{(n-1)^2} \left[\text{Var}(A) - \frac{2}{n} \text{Cov}(A, B^2) + \frac{1}{n^2} \text{Var}(B^2) \right]$$

Required Calculations

We need three quantities: $E(A^2)$, $E(AB^2)$, $E(B^4)$.

First Calculation
Since $A = \sum_{i=1}^n Y_i^2$,
$$A^2 = \left(\sum_{i=1}^n Y_i^2 \right)^2 = \sum_{i=1}^n Y_i^4 + 2 \sum_{i < j} Y_i^2 Y_j^2$$

Independence gives
$$E(A^2) = nE(Y_1^4) + 2 \binom{n}{2} \sigma^4 = nE(Y_1^4) + n(n-1)\sigma^4$$

Second Calculation
Because $B^2 = \left(\sum_{i=1}^n Y_i \right)^2 = \sum_{i=1}^n Y_i^2 + 2 \sum_{i < j} Y_i Y_j$, every term in the second part has at least one independent centered factor appearing in the first power, so its expectation is zero. Hence,
$$E(AB^2) = E(A^2) = nE(Y_1^4) + n(n-1)\sigma^4$$

The Fourth Moment of the Centered Sum

Recall that $B = \sum_{i=1}^n Y_i$, $E(B^2) = 0$.
When expanding B^4 , only terms in which every index appears at least twice have nonzero expectation.
Thus, the surviving terms are

$$\sum_{i=1}^n Y_i^4 \quad \text{and} \quad 6 \sum_{i < j} Y_i^2 Y_j^2$$

Therefore,
$$B^4 = \sum_{i=1}^n Y_i^4 + 6 \sum_{i < j} Y_i^2 Y_j^2 + \text{terms having expectation zero}$$

Taking expectations,
<